

A geometric reduced-order modeling for conductive phase-change materials

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Abstract

We develop a new reduced-order modeling (ROM) for phase-change material (PCM) process in a conductive regime with a moving solid-liquid interface. Guided by Stefan problem theory, we exploit the observed $\sqrt{t}$ scaling of the interface position to introduce a geometric reduction based on a self-similar coordinate transformation. Indeed such as transformation maps the physical, time-varying domain onto a reference ξ -grid and removes the moving discontinuity from the snapshot data. In fact the self-similar reformulation removes the dominant cost associated with representing front, allowing the reduced basis to focus on the diffusive layer's shape rather than its position. On top of this geometric reduction, we apply a ROM in the spirit of proper orthogonal decomposition (POD) based on ξ -grid, refer to GPOD. We assess this approach on a simplified one-dimensional model, obtained by averaging the temperature field along the y -axis, and subsequently on the full two-dimensional PCM problem. In both cases, GPOD achieves substantially faster convergence than classical POD applied to the physical temperature field: a 1-mode GPOD already captures 99.999% of the cumulative energy, and 4-mode reach 100%, whereas POD requires more than 20 modes to reach the same level of energy captured. Furthermore GPOD combined with radial basis function (RBF) establishes a non-intrusive low-rank efficient alternative to the conventional POD-Galerkin, which remains very prohibitive for phase-change heat transfer models.

Keywords: PCM, FEM, self-similar coordinate transformation, ROM, POD, RBF.

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1 Introduction

We address herein the solid-liquid transition driven by heat conduction, with an enthalpy source term encoding the latent heat absorbed at the melting front. This problem is a classical instance of the Stefan problem [Ste91, Rub71, AS93], in which a moving interface separates the solid and liquid phases and evolves according to a nonlinear coupling between the temperature field and the release/absorption of latent heat at the front. Rather than tracking explicitly the front, we adopt the fixed-domain *enthalpy formulation* [VC87, Cra84], which recasts the two-phase free-boundary problem as a single nonlinear diffusion equation posed on the whole domain Ω . The dynamics are then governed by the following heat transfer problem [Rub71, AS93]:

$$\partial_t T - \alpha \nabla^2 T = -\partial_t S(T), \quad \text{in } \Omega \times (0, t_{\max}), \quad (1)$$

$$T(\mathbf{x}, 0) = T_0, \quad \text{in } \Omega, \quad (2)$$

where T denotes the temperature field. S represents the enthalpy source term controlling the phase-change process [VC87, EHI⁺46], defined by

$$S(T) = \frac{1}{\text{Ste}} \mathbf{1}_{T \geq T_f}, \quad (3)$$

with $T_f = 0$ the fusion temperature, $\text{Ste} = 0.045$ the Stefan number, and $\alpha = 1/56.2$ the thermal diffusivity. The domain Ω is a square cavity, initially solid at uniform temperature $T_0 = -0.01$, and subsequently heated at the left wall at a fixed temperature $T = \mu \in [1.0, 1.10]$ above the fusion point, which triggers melting from that boundary [Gup03]. We define the moving interface

$$\Gamma(t) = \{\mathbf{x} \in \Omega : T(\mathbf{x}, t) = T_f\}, \quad (4)$$

where the source term $\partial_t S(T)$ concentrates as a moving singular (interfacial) contribution. Indeed, since $S(T)$ is simply $1/\text{Ste}$ times the characteristic function of the liquid region, its time derivative reduces, away from $\Gamma(t)$, to zero, while across $\Gamma(t)$ it recovers the classical Stefan condition. As the ratio $1/\text{Ste} \approx 22$ is large compared to α , the latent heat effect dominates over pure diffusion: the interface advances slowly relative to the thermal relaxation of each phase, and small errors in the localization of $\Gamma(t)$ can strongly affect the predicted melting dynamics. Accurate and stable numerical resolution of problem (1)–(2) is therefore required to correctly capture the position and velocity of the melting front, which motivates the numerical approach developed in this work.

PCM systems, in which a material undergoes a solid-liquid transition driven by thermal boundary conditions, are central to a wide range of engineering applications, including latent heat thermal energy storage, casting and solidification processes, and cryogenic systems [AS93]. The mathematical description of such systems belongs to the class of *Stefan problems*, characterized by a moving interface whose position is itself an unknown of the problem, coupled to the diffusion of temperature on either side [Cra84, Rub71]. While closed-form self-similar solutions exist for idealized one-dimensional, semi-infinite configurations [Rub71], realistic PCM systems generally require numerical treatment due to geometric complexity, finite domain size, and parameter dependence.

A standard numerical strategy for Stefan-type problems is the enthalpy formulation, which reformulates the moving-boundary condition as a fixed-domain nonlinear diffusion equation for the enthalpy field, thereby avoiding explicit front-tracking [VC87, AS93]. This formulation is well suited to conventional finite element discretizations and underlies many general-purpose PCM solvers, including the PCM numerical toolbox employed in this work [SRT⁺20]. However, despite its robustness, the enthalpy-based finite element approach remains computationally expensive when repeated simulations are required across a range of physical parameters, such as varying hot-wall temperatures, motivating the construction of reduced-order models (ROMs) capable of approximating the high-fidelity solution manifold at a fraction of the cost [BGW15, QMN16].

Among ROM techniques, POD is one of the most widely used approaches for constructing reduced bases from collections of high-fidelity snapshots, applied a SVD of the snapshot matrix [Sir87, BHL93]. POD has been successfully applied to a broad range of parametrized PDE problems [BGW15]; however,

its efficiency degrades significantly in the presence of *transport-dominated* or *moving-front* phenomena, where a coherent structure (a wave, shock, or interface) translates through the domain over time or across the parameter range. In such cases, a globally linear POD basis is known to converge slowly, since an increasing number of modes is required to reconstruct the translating feature at every possible front location [OR16, RML18]. This is precisely the difficulty encountered in PCM systems, where the solid-liquid interface propagates continuously through the domain as the simulation evolves.

Several strategies have been proposed in the literature to address this slow-decaying Kolmogorov n -width associated with transport-dominated problems, including registration-based and Lagrangian POD methods [RML18], shifted or transported snapshot approaches [OR16], and other nonlinear or manifold-based model reduction techniques. In the present work, we pursue a complementary strategy that is specifically tailored to the Stefan-problem structure of PCM systems. Motivated by the classical self-similar scaling of the interface position, $s(t) \sim \beta\sqrt{t}$, predicted by Stefan theory [Rub71, Cra84], we introduce a *geometric reduction* step prior to the algebraic reduction: a self-similar coordinate transformation $\xi = x/s(t)$ that maps the physical, time-varying domain onto a no-uniform reference ξ -grid. This parametrized ξ -grid plays a central role in the reduction strategy, as it explicitly separates the two sources of variability present in the snapshot data: the translation of the front and the evolving shape of the diffusive thermal layer. Because the coordinate transformation is built directly from the physics of the problem, the resolution and placement of the ξ -grid points directly control the accuracy with which the interpolation steps ($T(x) \rightarrow X(\xi)$ and back) preserve the sharp gradients near the interface, making the choice of ξ -grid a key ingredient governing the overall approximation accuracy of the reduced-order model.

However the GPOD does not easily support a Galerkin-projection-based online stage, unlike classical POD, which is naturally defined on the same finite element space as the high-fidelity solver and can therefore be projected directly onto the governing weak form (1). Because the GPOD basis $\varphi_k(\xi)$ is instead defined on the parametrized ξ -grid, which differs from the reference finite element space and evolves implicitly through the time-dependent map $\xi = x/s(t)$, a standard Galerkin projection of the governing equations onto this basis is not straightforward: the spatial operators (stiffness, mass matrices) of the finite element discretization are not directly expressible in the ξ -coordinate without re-deriving the weak form under the moving coordinate change, which reintroduces additional nonlinear and time-dependent terms. This makes GPOD less directly amenable to intrusive, physics-based online reduced-order solvers than classical POD. Radial basis function (RBF) interpolation-based surrogate modeling [Buh03] appears to be a more natural alternative in this setting: rather than seeking a Galerkin projection of the governing equations, an RBF surrogate can be trained directly on the GPOD coefficients as a function of the physical parameters and time, providing a fully non-intrusive, low-rank surrogate for the PCM model that sidesteps the need to reformulate the finite element operators on the ξ -grid.

Applying standard POD to the resulting self-similar snapshots, rather than to the original physical-space snapshots, yields what we refer to as the *geometric POD* (GPOD). Because the moving discontinuity has already been removed by the geometric transformation, the GPOD basis is expected to converge substantially faster with a few retained modes than classical POD applied directly to the physical temperature field. This offers a robust no-intrusive RBF time dynamic surrogate for the latent POD coefficient. The goal of this paper is to quantify this gain, first in a simplified one-dimensional setting obtained by averaging the two-dimensional temperature field along the direction transverse to the dominant heat flux, and subsequently in the full two-dimensional PCM problem.

The remainder of the paper is organized as follows. Section 2.1 introduces the geometric reduction including interface-tracking procedure and the self-similar coordinate transformation. Section 2.2 describes shortly the reduced-order modeling reviewing the POD approach and the RBF interpolation technique. Section 3 reports the numerical comparison between GPOD and POD, first for the simplified one-dimensional averaged model and then for the full two-dimensional PCM problem, followed by concluding remarks in Section 4.

2 Modeling

In the absence of convection, the cavity is initially filled with solid material at a uniform temperature of $T = -0.01$. Heat is then imposed at the left wall, initiating the melting process and since the dynamic is purely diffusive, the liquid–solid interface progressively propagates from left to right. Due to the natural conduction, the heat transfer is predominantly one-dimensional, occurring mainly along the horizontal direction, while temperature variations along the vertical direction remain weak. Consequently, the temperature field can be approximated by its vertical average,

$$\bar{T}(x, t) = \frac{1}{N_y} \sum_{i=1}^{N_y} T(x, y_i, t). \quad (5)$$

2.1 Geometric reduction

In this section we present our geometric reduction modeling aimed to model the moving liquid–solid interface and then aligned the temperature $\bar{T}$ at a single static front thanks to the self-similar variable change.

Front modeling As mentioned above, the liquid–solid thin interface remains nearly vertical throughout the melting process, indicating that its horizontal position varies only marginally with the vertical coordinate. The moving interface is therefore characterized by

$$\Gamma_s = \{(x_s, t) \mid \bar{T}(x_s, t) = T_m\},$$

where T_m is the melting temperature at the interface set to $T_m = 10^{-2}$.

According to the Stefan condition [Cra84, Rub71] that assumes an approximately uniform temperature gradient within the liquid region, the interface position may be approximated by the following nonlinear law,

$$s(t) = x_0 + \beta\sqrt{t}, \quad (6)$$

where the parameters x_0 and β are identified through least-squares fitting [PTVF07] using the detected interface positions $\{(x_s^n, t^n)\}_{n=1}^{N_t}$. While based on the the Stefan assumption, the theoretical value of β is given by

$$\beta_{\text{theory}} = \sqrt{2 \text{Ste} \alpha}. \quad (7)$$

Self-similar transformation Motivated by the previous observations and the classical Stefan solution [AS93, Cra84], the temperature field is assumed to satisfy the self-similar form

$$\bar{T}(x, t) \approx f\left(\frac{x}{\sqrt{t}}\right). \quad (8)$$

Accordingly, we introduce the self-similar coordinate

$$\xi_t = \frac{x}{s(t)},$$

and define the transformed temperature field

$$\tilde{X}(\xi_t) = \bar{T}(x, t), \quad x \in [0, 1], \quad t \in [2.1, 50.6].$$

Since the interface position evolves in time, the coordinate ξ_t is also time dependent. Consequently, all snapshots are first interpolated onto a fixed ξ -grid before being mapped back to the physical domain. The procedure consists of backward and forward transformations as pictured in the Figure 1 below: The choice of the parametrized ξ -grid is crucial for the accuracy of the method and needs careful attention. Indeed a uniform discretization generally introduces interpolation errors that may offset the advantages of the self-similar representation:

$$X(\xi, t) \approx \bar{T}\left(\frac{x}{s(t)}, t\right).$$

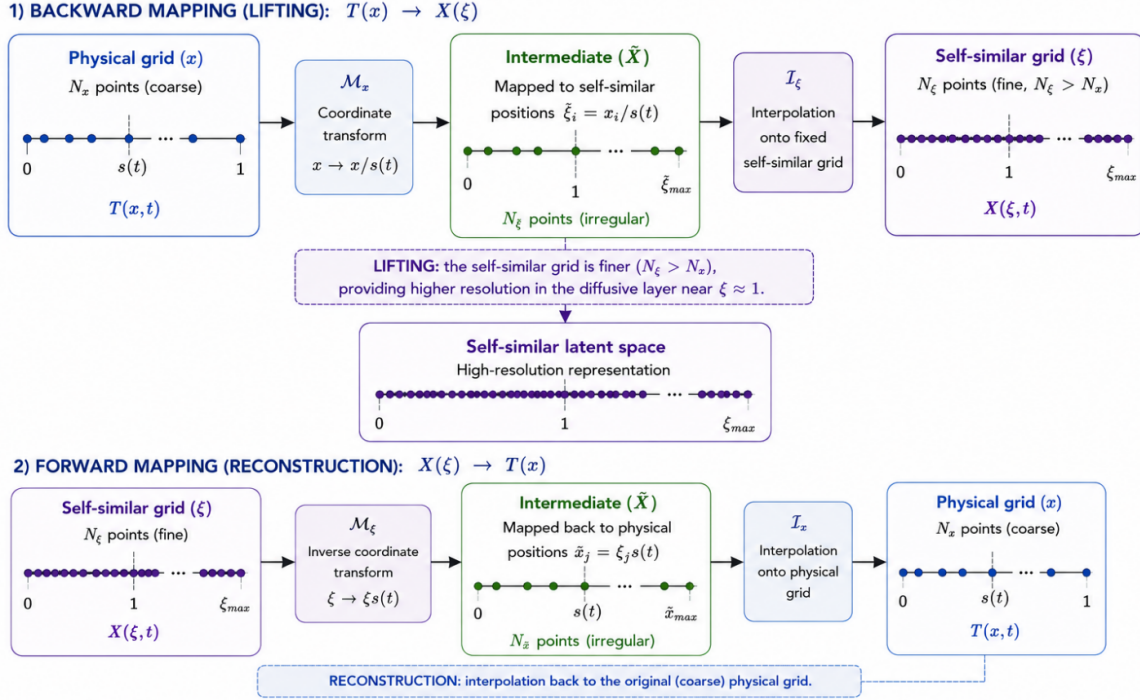


Figure 1: Schematic of the self-similar snapshots processing: first the backward step lifts the snapshots to higher self-similar grid resolution then the forward mapping interpolate from the self-similar latent space into the physical grid.

Because the temperature field exhibits only weak dependence on the vertical coordinate, the one-dimensional self-similar formulation naturally extends to two dimensions by preserving the y -coordinate,

$$X(\xi, y, t) \approx T\left(\frac{x}{s(t)}, y, t\right).$$

This self-similar formulation constitutes the basis of the reduced-order model developed in the following section.

2.2 Reduced order modeling

Let $\{T(x, t_n)\}_{n=1}^N$ denote the snapshot matrix $T \in \mathbb{R}^{N \times N_h}$, where N_h is the number of degrees of freedom of the spatial mesh and N the number of snapshots.

Proper Orthogonal Decomposition (POD) The low-rank POD method [Sir87, BHL93] seeks a low-dimensional basis $\{\phi_k\}_{k=1}^r \subset \mathbb{R}^{N_h}$ such that the r -rank approximation

$$T_r(x, t_n) = \sum_{k=1}^r a_k(t_n) \phi_k(x), \quad (9)$$

minimizes the total reconstruction error $\|T(x, t) - T_r(x, t)\|_2$. The modes ϕ_k and coefficients $a_k(t_n)$ are obtained via the Singular Value Decomposition (SVD) [Vol11] :

$$T = U \Sigma V^\top, \quad (10)$$

where $U \in \mathbb{R}^{N \times N}$ and $V \in \mathbb{R}^{N_h \times N_h}$ are orthogonal matrices and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_N)$ with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ are the decreasing singular values. The POD modes are the columns of V :

$$\phi_k = V_{:,k}, \quad k = 1, \dots, r, \quad (11)$$

and the latent POD coefficients are given by

$$a_k(t_n) = \phi_k T(x, t_n), \quad (12)$$

where the dimension r are defined such that to capture 100% of the cumulative energy ration:

$$\mathcal{E}(r) = \frac{\sum_{k=1}^r \sigma_k^2}{\sum_{k=1}^N \sigma_k^2}. \quad (13)$$

In practice r is easily truncated by

$$\mathcal{E}(r) \geq 1 - \epsilon,$$

where the tolerance ϵ is small as possible to guarantee sufficient approximation accuracy while maintaining a compact reduced basis.

POD-Galerkin Beyond the query parameters used at the ROM decomposition, the representation (12) is no longer valid and thus conventional Galerkin projection method is mostly used to compute coefficient at target parameter. The approach consists in projecting the full-order problem (1) onto the reduced space $V_r = \text{span}\{\phi_1, \dots, \phi_r\}$. Then substituting T_r (9) in the following weak formulation

$$(\partial_t T_r, \phi_i) + \alpha(\nabla T_r, \nabla \phi_i) = -(\partial_t S(T_r), \phi_i), \quad \forall \phi_i \in V_r, \quad (14)$$

provides equation to get α_r . This equation may be computationally expensive during the online phase, as its evaluation requires solving a nonlinear system and involves physical quantities that may not be readily reduced.

POD-RBF In contrast to the intrusive POD-Galerkin approach, the unsteady coefficient a_r are defined with respect to the query time-parameter pair (t, μ) . As already proposed in [XFP15], the RBF interpolation is employed to construct a surrogate model in offline. During this stage, the reduced basis and the kernel interpolator are constructed from the snapshot data where the RBF interpolator $\mathcal{I}_{\text{RBF}}$ is explicitly defined as:

$$\mathcal{I}_{\text{RBF}}^i(t, \mu) = \sum_{m=1}^N w_{i,m} \varphi(\|(t, \mu) - (t_m, \mu_m)\|), \quad i = 1, \dots, r, \quad (15)$$

where the radial function φ may take one of the forms:

$$\text{Cubic: } \varphi(\|(t, \mu)\|) = \|(t, \mu)\|^3, \quad \text{Spline: } \varphi(\|(t, \mu)\|) = \|(t, \mu)\|^2 \log(\|(t, \mu)\|).$$

The RBF weights $w_{i,m}$ are computed once by enforcing the interpolation condition:

$$\sum_{m=1}^N w_{i,m} \varphi(\|(t_n, \mu_n) - (t_m, \mu_m)\|) = \phi_i(x) T(x, t_n, \mu_n), \quad n = 1, \dots, N.$$

This yields to solve once r linear-independent system of equations of dimension N that easy to parallelize. Finally for any parameter μ_* , the surrogated POD-RBF model holds:

$$T_r(x, t, \mu_*) = \sum_{k=1}^r \mathcal{I}_{\text{RBF}}^k(t, \mu_*) \phi_k(x), \quad (16)$$

3 Numerical results

Conventional finite element method is applied to solve the problem (1), following the implementation provided in the PCM numerical toolbox [SRT+20].

High-fidelity snapshots are generated for several values of the hot-wall temperature parameter $\mu \in [1, 1.1]$. Each simulation is carried out over the time interval $[2.1, 51.6]$ using a uniform time step $\delta t = 0.5$, resulting in $N_t = 100$ time steps per trajectory. The spatial discretization uses a fixed, non-uniform square mesh of size 60×60 with P2 finite elements.



Figure 2: High fidelity temperature at time $t = 2.1$ (left), $t = 27.6$ (middle) and the final time $t = 51.6$ (right). Blue denotes the solid phase ($T < 0$), whereas red denotes the liquid phase ($T \geq 0$). Intermediate colors represent the thermal transition layer that develops as heat diffuses into the material.

Figure 2 illustrates snapshots of the temperature field at different simulation times for fixed $\mu = 1$. As shown, the conductive heat transfer exhibits quasi-uniform diffusion along the y -axis, as expected, while the liquid-solid interface remains vertical throughout the entire simulation. This behavior supports the 1D reduction and the self-similar modeling investigated in the next sections.

In what follows, we continue the study of two reduced-order models: the standard POD, applied directly to the original snapshot matrix T , and the GPOD, referring to POD applied to the self-similar snapshot matrix X . We first analyze the self-similar modeling for the 1D case, and then turn to the full 2D PCM problem.

3.1 1D self-similar modeling

We start by validating the self-similar variable change for single parameter μ to highlight the benefit of such transformation. The parametrized ξ -grid holds non-uniform grid over $\xi \in (0, 2.5)$, allocating the most grid points to the interval $(0, 1.095)$ arising the diffusion and less points on the static part $(1.095, 2.5)$. In Figure 3 we plot two snapshots on physical grid and then on the self-similar ξ -grid. As expected each snapshot has its own interface position that demonstrates the moving liquid-solid

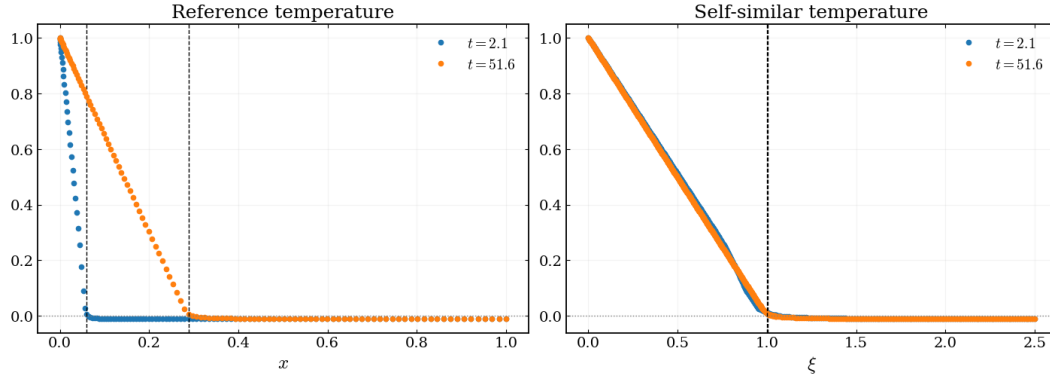


Figure 3: Reference and self-similar temperatures at time $t = 2.1$ and $t = 51.6$ obtained with parameter $\mu = 1$.

interface on the x -grid while both isolines remain very closed on one interface in the ξ -grid around $\xi = 1$. This result confirms a good agreement with the physical theory and provides efficient reduced-order modeling in the following section. Next we assess the accuracy of the coordinate transformation by measuring the interpolation error

$$\mathcal{I}_\infty = \frac{\max_n |\bar{T}^n - \mathcal{F}(X^n)|}{\max_n |\bar{T}^n|}, \quad \mathcal{I}_2 = \frac{|\bar{T} - \mathcal{F}(X)|_2}{|\bar{T}|_2}$$

Table 1: Errors $\mathcal{I}_t$ for different resolutions N_ξ at selected snapshots.

N_ξ	$\mathcal{I}_\infty$	$\mathcal{I}_2$
121	9.33×10^{-5}	8.25×10^{-6}
242	2.25×10^{-5}	1.59×10^{-6}
484	3.88×10^{-6}	1.02×10^{-6}
605	3.88×10^{-6}	1.01×10^{-6}

with $X = \mathcal{B}(\bar{T})$ where $\mathcal{F}$ and $\mathcal{B}$ denote the backward and forward interpolation operators between the physical and self-similar grids. Table 1 reports maximum relative absolute and L^2 errors which remain below 10^{-5} for the finest grid, demonstrating that the coordinate transformation introduces only a small numerical perturbation compared with the reference temperature.

3.2 1D reduced-order modeling

Knowing the self-similar snapshot matrix X , we can now compare the POD and GPOD reduced-order modeling using the same reduced dimension r . Table 2 reports the cumulative energy $\mathcal{E}(r)$ defined in

Table 2: Cumulative energy $\mathcal{E}(r)$ (%) for POD and GPOD.

Modes r	ROM(T)	ROM(X)
1	97.5093	99.9991
2	99.7081	99.9998
3	99.9305	99.9999
4	99.9756	100.000
10	99.9990	100.000
20	99.9999	100.000

Eq. (13) for both reduced-order models. Obviously the self-similar formulation exhibits remarkable energy compaction: a single ROM(X) mode captures more than 99.99% of the total energy, while four modes recover virtually 100%. This behavior is consistent with the low-rank structure predicted by the self-similar hypothesis in Eq. (8) and agrees with observations reported in related studies [RSSM18]. By contrast, the ROM($\bar{T}$) applied to the original temperature field requires more than 20 modes to achieve a comparable level of energy captured. To further assess the performance of both approaches, we measure the relative reconstruction error,

$$\varepsilon_r(Y) = \frac{|Y - Y_r|_2}{|Y|_2}, \quad (17)$$

reported in Table 3. This Table highlights more the superiority of the self-similar reduced-order model,

Table 3: Relative reconstruction error ε_r .

Modes r	$\varepsilon_r(T)$	$\varepsilon_r(X)$
1	1.56×10^{-1}	2.93×10^{-3}
2	5.40×10^{-2}	1.41×10^{-3}
3	2.64×10^{-2}	8.71×10^{-4}
4	1.56×10^{-2}	4.59×10^{-4}
10	3.14×10^{-3}	5.49×10^{-5}
20	8.35×10^{-4}	3.60×10^{-6}

with only one POD mode, GPOD achieves a relative reconstruction error of magnitude 10^{-3} , whereas the conventional POD requires about ten modes to reach a similar level of accuracy. Moreover, a 4-mode GPOD outperforms a 20-mode POD, yielding a more accurate reconstruction of the temperature field while using only a fraction of the degrees of freedom. These results clearly demonstrate the substantial reduction in dimensionality achieved by the self-similar formulation without compromising reconstruction accuracy. Figure 4 shows an excellent agreement between the reduced temperature

obtained by 1-mode GPOD with the reference FOM temperature. While the POD produces a poor reconstruction for both examples with wrong liquid-solid interface.

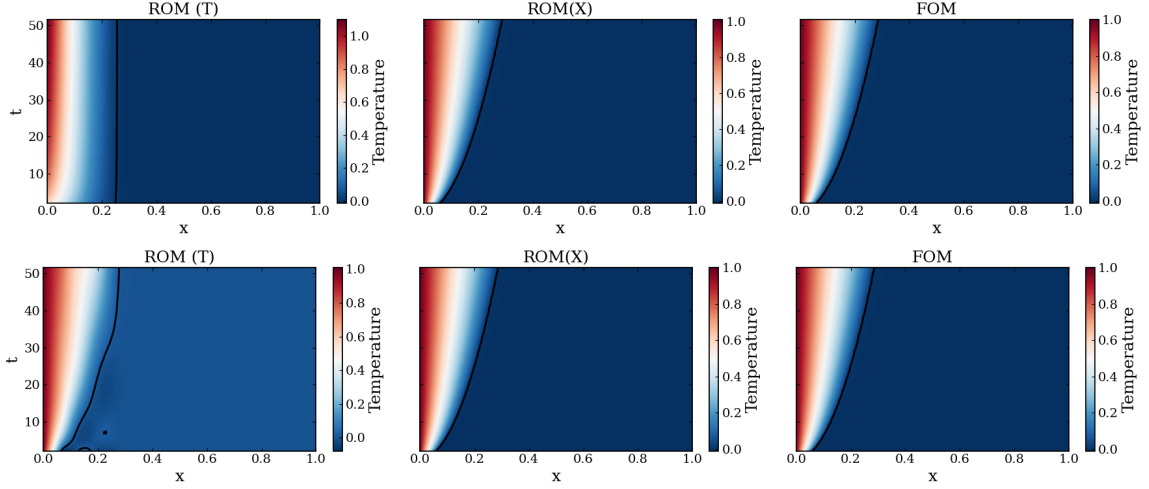


Figure 4: ROM reconstructions for the reference and self-similar snapshots compared to the FOM obtained by 1-mode (top) and 4-mode (bottom) with the moving liquid-solid interface in black.

3.3 2D reduced-order modeling

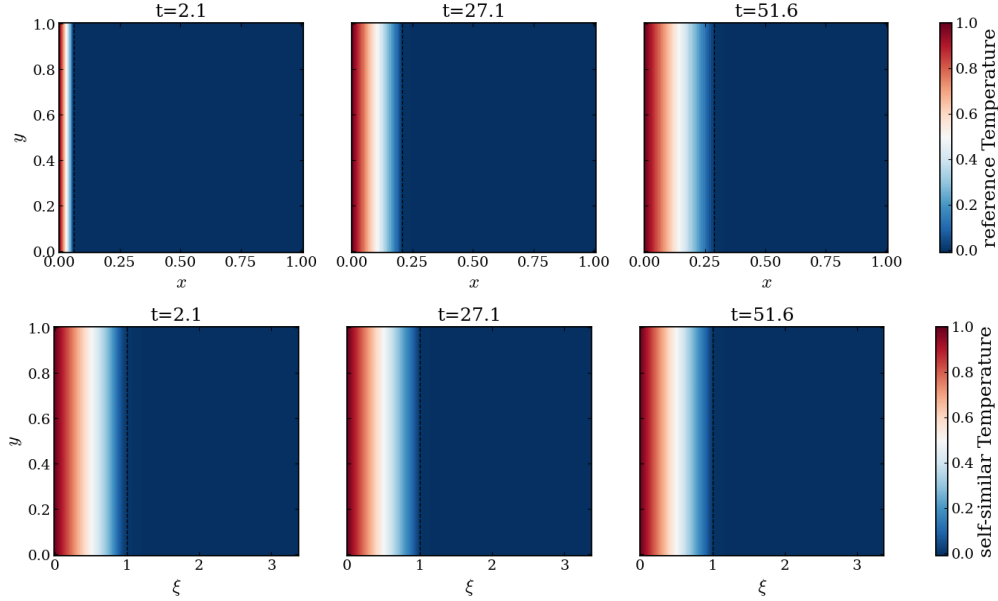


Figure 5: Reference and self-similar snapshots respect to (x, y) grid (top) and (ξ, y) grid (bottom) for $\mu = 1$ with the moving front (dash line).

After analyzing the one-dimensional reduced-order model, we now evaluate the GPOD methodology within a two-dimensional framework. Under the assumption that temperature varies weakly along the vertical, the self-similar snapshot tensor $X \in \mathbb{R}^{N_t \times N_\xi \times N_y}$ is derived from the physical temperature field via the coordinate transformation:

$$X(\zeta, y, t) \approx T(x/s(t), y, t).$$

Figure 5 compares three representative snapshots in both the physical domain and the self-similar coordinate system. In the physical domain, the temperature field T exhibits a pronounced translation

of the liquid–solid interface as melting progresses. Conversely, following the self-similar coordinate transformation, the interface is effectively aligned around $\xi = 1$, producing snapshots of X that are nearly indistinguishable. This alignment significantly reduces the apparent temporal variability of the solution, revealing its intrinsic low-rank structure and making the self-similar representation particularly well-suited for ROM construction.

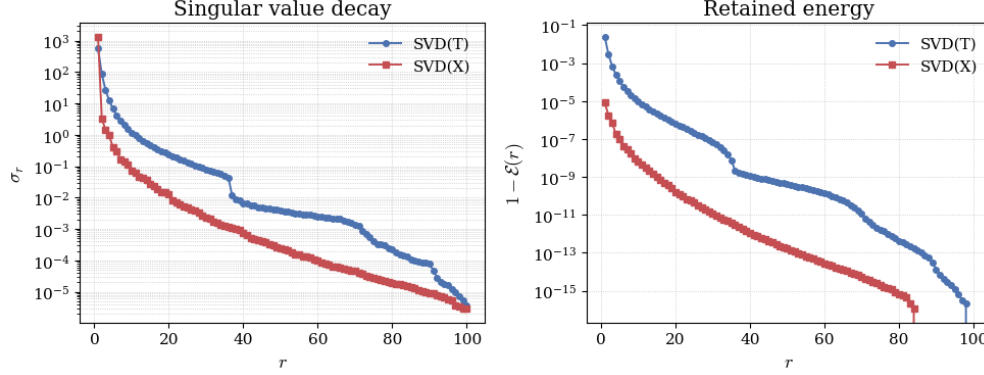


Figure 6: Comparison of SVD(T) and SVD(X): left panel is the singular value decay and right panel is truncated cumulative energy $1 - \mathcal{E}(r)$

3.3.1 POD and GPOD

We now turn to the application of ROMs to both original temperature field T and aligned snapshot X using standard SVD technique. Figure 6 illustrates the resulting singular value decay and retained energy as a function of the rank r . As shown, SVD(X) exhibits a rapid initial decay in singular values for $r \leq 20$, beyond which it decays slowly, mirroring the behavior of SVD(T). Regarding retained energy, GPOD achieves a low-rank, compact reduced basis within a reasonable energy tolerance range of $\epsilon \in [10^{-9}, 10^{-5}]$, whereas standard POD requires a significantly larger energy threshold ($\epsilon \leq 10^{-9}$) to yield an accurate basis. Table 4 clearly demonstrates the superiority of GPOD over standard POD

Table 4: Comparison of the cumulative energy $\mathcal{E}(r)$ and relative reconstruction error ε_r .

Modes r	POD		GPOD	
	$\mathcal{E}(r)$ (%)	$\varepsilon_r(T)$	$\mathcal{E}(r)$ (%)	$\varepsilon_r(X)$
1	97.5094	1.57×10^{-1}	99.9992	2.85×10^{-3}
2	99.7082	5.40×10^{-2}	99.9998	1.34×10^{-3}
3	99.9305	2.64×10^{-2}	99.9999	8.30×10^{-4}
4	99.9756	1.56×10^{-2}	100.0000	4.28×10^{-4}
10	99.9990	3.10×10^{-3}	100.0000	8.04×10^{-5}
20	99.9999	8.10×10^{-4}	100.0000	1.25×10^{-5}

in terms of both cumulative energy and reconstruction error. Even with a single mode ($r = 1$), GPOD captures more than 99.999% of the snapshot energy, compared to only 97.509% for conventional POD. This remarkable energy compaction translates directly into a substantially lower reconstruction error: the 1-mode GPOD achieves a relative error of 2.85×10^{-3} , whereas POD yields an error of 1.58×10^{-1} . Furthermore GPOD converges toward the ground truth faster than the POD, requires only fewer mode. For example, a 3-mode GPOD achieves a comparative reconstruction error of just $8. \times 10^{-4}$, against 20-mode standard POD model ($8. \times 10^{-4}$). These results confirm that aligning the moving liquid–solid interface via the self-similar transformation dramatically enhances the low-rank structure of the snapshot manifold, allowing highly accurate reduced-order models to be constructed with very few basis functions.

3.3.2 Consistency of GPOD with the Physical Problem

While the convergence of the cumulative energy $\mathcal{E}(r)$ and relative reconstruction error ε_r quantify how well the GPOD and POD bases approximate the snapshot data, they do not guarantee consistency with the underlying governing physics, namely, that the reduced approximation satisfies the underlying heat transfer equation (1). This distinction is particularly critical for GPOD because its reduced basis is constructed in self-similar coordinates ξ , which differ from the physical coordinates in which the governing PDE and its finite element discretization are posed. To evaluate physical consistency, we introduce the following diagnostic metrics:

$$\begin{aligned}\mathcal{I}_\infty(Y) &= \frac{\max_n |T_h^n - Y^n|}{\max_n |T_h^n|}, \\ \mathcal{E}_2^2(Y) &= \frac{1}{N_t} \sum_{n=1}^{N_t} \frac{(T_h^n - Y^n, T_h^n - Y^n)_{L^2(V_h)}}{(T_h^n, T_h^n)_{L^2(V_h)}}, \\ \mathcal{R}_2^2(Y) &= \frac{1}{N_t} \sum_{n=1}^{N_t} (R^n, R^n)_{L^2(V_h)},\end{aligned}$$

with

$$(R^n, v)_{L^2(V_h)} = (\partial_t Y^{n+1}, v)_{L^2(V_h)} + \alpha(\nabla_h Y^{n+1}, \nabla_h v)_{L^2(V_h)} + (S(Y^{n+1}), v)_{L^2(V_h)}, \quad v \in V_h,$$

where V_h represents the reference finite element space and $(\cdot, \cdot)_{L^2(V_h)}$ denotes the standard L^2 inner product on V_h .

Denoting by $T_s = \mathcal{F}(\mathcal{B}(T_h))$ as the self-similar backward and forward transformation mapped back to the reference T_h , we evaluate these metrics in Table 5. The resulting errors and residuals decrease

Table 5: Errors and residuals for the geometric snapshot transformation at different resolutions N_ξ .

N_ξ	$\mathcal{I}_\infty(T_s)$	$\mathcal{E}_2(T_s)$	$\mathcal{R}_2(T_s)$
121	2.15×10^{-4}	1.13×10^{-5}	8.63×10^{-3}
242	5.37×10^{-5}	2.77×10^{-6}	1.21×10^{-3}
484	7.46×10^{-7}	2.15×10^{-6}	1.70×10^{-4}
605	1.77×10^{-7}	2.14×10^{-6}	8.75×10^{-5}

with respect to the ξ -grid resolution, reaching a residual error of 8.75×10^{-5} . These results confirm that the self-similar mapping preserves key physical features. Next, for the ROM investigation, we

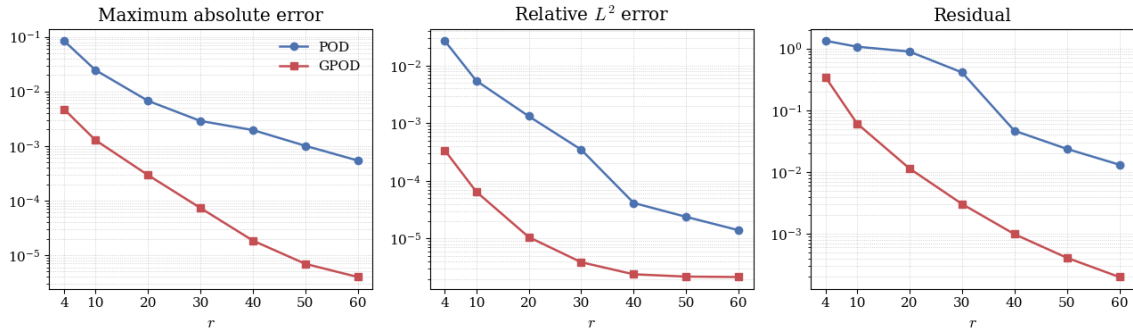


Figure 7: Errors and residual obtained by GPOD and POD respect to the truncated mode r .

fix $N_\xi = 605$. Figure 7 presents a comparison of $\mathcal{I}_\infty(T_r)$, $\mathcal{E}_2(T_r)$, and $\mathcal{R}_2(T_r)$ for reduced states T_r generated by both GPOD and POD. At lower mode counts ($r = 4, 10, 20, 30$), GPOD exhibits smaller values across all three metrics, demonstrating higher accuracy than standard POD. However, beyond

$r = 40$, the GPOD's metric $\mathcal{E}_2(T_r)$ saturates near $\mathcal{E}_2(T_s)$, while POD quantities continue to improve over this higher-rank range. This saturation is an intrinsic consequence of the spatial interpolation required on the ξ -grid: because every GPOD snapshot undergoes a backward mapping $T(x) \rightarrow X(\xi)$ followed by a forward reconstruction $X(\xi) \rightarrow T(x)$, the interpolation error introduced by this round-trip transformation establishes an accuracy floor that cannot be lowered by adding more POD modes. In other words, GPOD's ultimate accuracy is bounded below by the discretization error of the self-similar transformation. Once the POD truncation error falls below this floor, adding further modes yields no additional overall improvement. Interestingly, GPOD maintains a lower maximum pointwise error ($\mathcal{I}_\infty$) than POD even up to $r = 60$. This indicates that while its global L^2 error and residual consistency plateau, GPOD remains locally more accurate—particularly near the sharp gradient at the moving interface around $\xi = 1$.

3.3.3 POD-RBF Interpolation

In this last section, we assess the non-intrusive RBF-based time-and-parameter surrogate modeling, combined with both classical POD and GPOD. The training time interval remains unchanged, while the hot-wall temperature parameter is sampled over the discrete set $\{1, 1.02, 1.04, 1.06, 1.08, 1.10\}$. The RBF interpolation then provides a surrogate model over the temporal and parametric dimensions, while the spatial dependence is carried by the reduced-order basis, according to:

- **POD-RBF:**

$$T_r(\mathbf{x}, t, \mu) = \sum_{k=1}^r \mathcal{I}_{\text{RBF}}^k(t, \mu) \phi_k(\mathbf{x}), \quad (18)$$

- **GPOD-RBF:**

$$T_r(\mathbf{x}, t, \mu) = \mathcal{F} \left(\sum_{k=1}^r \mathcal{I}_{\text{RBF}}^k(t, \mu) \psi_k(\xi) \right), \quad (19)$$

where $\mathcal{I}_{\text{RBF}}^k(t, \mu)$ denotes the k -th RBF-interpolated temporal/parametric coefficient, ϕ_k and ψ_k are the spatial POD and GPOD modes, respectively. $\mathcal{F}$ is the geometric mapping that transforms the self-similar reconstruction back into the physical domain. Each decomposition retains its own truncation rank r and its own RBF interpolator, both trained over the same time and parameter sampling.

Figure 8 shows the complete offline training history for the RBF stage, for both the POD and GPOD low-rank models.

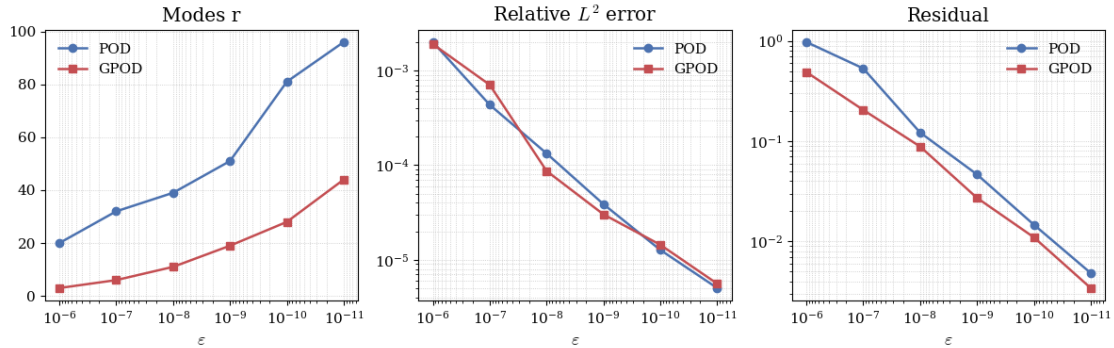


Figure 8: Training history of the RBF interpolation for POD and GPOD, with respect to the energy tolerance ϵ : left – retained rank r ; center – maximum of the relative L^2 error $\mathcal{E}_2$; right – maximum of the residual $\mathcal{R}_2$.

From this figure, we first observe that, for a given energy tolerance ϵ – in particular for $\epsilon = 10^{-8}$ GPOD reaches the target tolerance with a noticeably smaller rank $r = 11$ than classical POD $r = 39$, which confirms the benefit of the self-similar transformation. At the strictest tolerance considered, $\epsilon = 10^{-11}$, POD and GPOD retain 96 and 44 modes, respectively, corresponding to only 16% and 7% of the total number of snapshots.

However, regarding the $\mathcal{E}_2$ error metric, classical POD outperforms GPOD at this high-rank end of the spectrum, since GPOD relies on a reduced basis that is intrinsically less rich than that of POD at high rank. As discussed in Section 3.3.2, the main reason for this is that the geometric transformation itself absorbs a residual interpolation error, which limits how far the low-rank efficiency of GPOD can be pushed. Both POD and GPOD nonetheless exhibit monotonically decreasing errors and residuals as ϵ is tightened, confirming that both reduced-order surrogates remain consistent with the underlying physical problem.

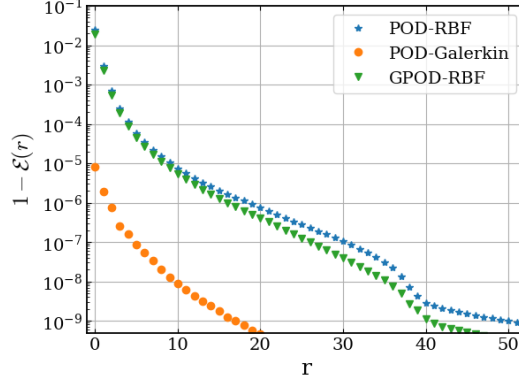


Figure 9: Retained energy ($1 - \mathcal{E}(r)$) with respect to the reduced dimension r for POD-RBF, POD-Galerkin, and GPOD-RBF

Finally, we evaluate the predictive performance of the POD-RBF and GPOD-RBF reduced-order models by choosing a energy threshold tolerance $\epsilon = 10^{-9}$. At this level, the intrusive POD-Galerkin holding 42 modes, as illustrated in Figure 9, is comforted against both non-intrusive RBF ROMs. Table 6 reports the resulting metrics, including the relative approximation errors and residuals for each ROM. Overall, the GPOD-RBF surrogate achieves approximation errors and residuals of the same

Table 6: Rollout metrics for non-intrusive and intrusive ROMs with respect to the parameter μ .

ROM	$\mu = 1.01$		$\mu = 1.05$		$\mu = 1.09$	
	$\mathcal{E}_2 \times 10^{-4}$	$\mathcal{R}_2 \times 10^{-2}$	$\mathcal{E}_2 \times 10^{-4}$	$\mathcal{R}_2 \times 10^{-2}$	$\mathcal{E}_2 \times 10^{-4}$	$\mathcal{R}_2 \times 10^{-2}$
POD-Galerkin	1.10	6.17	0.71	6.51	0.88	7.43
POD-RBF	1.45	4.47	0.65	4.63	1.34	4.92
GPOD-RBF	1.77	4.81	0.95	3.18	1.39	4.97

order of magnitude as those obtained with the POD-Galerkin and POD-RBF approaches. Despite this comparable level of accuracy, GPOD-RBF provides a more compact representation and constitutes the lowest-dimensional ROM among the three approaches. These results demonstrate that GPOD-RBF can retain the predictive accuracy of the conventional reduced-order models while significantly reducing the complexity of the reduced representation.

4 Conclusion

This work investigated a ROM strategy for conductive phase-change material systems, combining a physics-informed geometric reduction, based on a self-similar coordinate transformation, with a standard POD reduction. The resulting geometric-POD was compared against classical non-intrusive POD-RBF and conventional POD-Galerkin applied directly to the physical temperature field. The main benefit of GPOD is clearly established by our numerical studies: by removing the moving discontinuity from the snapshot data prior to the algebraic reduction step, GPOD yields a remarkably low-rank and compact basis, capturing 99.999% of the cumulative energy with a single mode and reaching 100% energy captured with only four modes, whereas classical POD requires more than twenty modes to achieve a comparable level of energy capture. This confirms that GPOD is accurate already

at very few modes, in sharp contrast to classical POD, which only becomes competitive at considerably higher rank once enough modes have been accumulated to compensate for the unresolved front translation.

These benefits, however, come with two notable drawbacks, both directly tied to the parametrized ξ -grid underlying the geometric reduction step.

1. **Dependence on the parametrized ξ -grid.** Unlike classical POD, which operates directly on the physical snapshots without any auxiliary transformation, GPOD requires the construction of a self-similar coordinate grid $\xi = x/x_f(t)$, together with the associated backward and forward interpolation steps $T(x) \rightarrow X(\xi) \rightarrow T(x)$. The resolution, placement, and interpolation scheme chosen for this ξ -grid are not free of approximation error, and directly affect the accuracy that GPOD can ultimately achieve. This introduces an additional modeling ingredient, and hence an additional source of tunable error, that is entirely absent from the classical POD pipeline.
2. **Saturation of the error.** Our results show that, beyond approximately $r = 40$ retained modes, the relative L^2 reconstruction error $\mathcal{E}_2$ of GPOD stop decreasing and instead saturate at a floor level. This saturation is a direct consequence of the ξ -grid interpolation itself: since every GPOD snapshot is first mapped through the backward step $T(x) \rightarrow X(\xi)$ and later reconstructed through the forward step $X(\xi) \rightarrow T(x)$, the interpolation error introduced by these two steps, evaluated *without any POD truncation* (i.e., on the exact self-similar snapshots themselves), constitutes a floor that no amount of additional POD modes can reduce further. In other words, the accuracy of GPOD is fundamentally bounded below by the accuracy of the self-similar snapshot transformation, and once the POD truncation error drops below this floor, adding further modes no longer improves the reduced-order approximation. As a result, the GPOD error for $r \gtrsim 40$ is, by construction, always less than or equal to the forward/backward interpolation error of the bare coordinate transformation, but it cannot be driven arbitrarily close to zero the way classical POD error can, in principle, by increasing r up to the full snapshot rank.

In summary, these findings position GPOD as a highly efficient low-rank alternative to classical POD for phase-change heat-transfer problems, particularly in the low-to-moderate rank regime that is most relevant to practical ROM applications. In this setting, retaining tens of POD modes may become computationally prohibitive, whereas GPOD can achieve comparable accuracy with a substantially more compact representation. Furthermore, when combined with RBF interpolation, GPOD-RBF provides a non-intrusive, low-rank, and computationally efficient surrogate model. This approach avoids the difficulties associated with formulating and implementing an intrusive Galerkin projection while retaining a competitive level of accuracy. Future work will investigate adaptive ξ -grid constructions to further improve the self-similar representation and mitigate the observed saturation effect.

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